

Jubair Ahmed Nabin

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Summary

Lecturer and researcher in Computer Science & Engineering at IUBAT, Dhaka. Research focuses on federated learning, intrusion detection, explainable AI, adversarial machine learning, and video anomaly detection. Author of 8+ conference papers at IEEE, Springer, and other venues.

Education

B.Sc. Chittagong University of Engineering & Technology (CUET), Computer Science and Engineering 2018 – 2023

- CGPA: 3.64 / 4.00
- Relevant Coursework: Machine Learning, Data Structures & Algorithms, Computer Networks, Database Management Systems, Artificial Intelligence, Software Engineering

Experience

International University of Business Agriculture and Technology (IUBAT), Lecturer, Department of CSE Dhaka, Bangladesh
Jan 2023 – present
3 years 9 months

- Courses taught: Data Structures, Database Management Systems, System Analysis & Design, Engineering Ethics, Fundamentals of Computing, Microsoft Office/Excel
- Developed lab materials and assessment rubrics for DBMS and Data Structures courses
- Mentored undergraduate students in research projects leading to conference submissions

IUBAT / CUET, Independent Researcher Dhaka, Bangladesh
Jan 2023 – present
3 years 9 months

- Designed and implemented a CNN-LSTM federated learning framework for network intrusion detection, preserving privacy across distributed clients while maintaining competitive detection accuracy
- Developed a two-stage video anomaly detection pipeline combining CNNs and vision transformers for temporal-spatial anomaly localization
- Investigated cross-domain adversarial robustness (PGD and C&W attacks) and SHAP-based explainability stability in phishing detection systems
- Built a multimodal (text + image) meme sentiment analysis system using XAI methods; extended to procrastination level assessment via ML and SHAP-based behavioral feature analysis

Publications

Federated Deep Learning for Cybersecurity and Intrusion Detection in Decentralized Networks Jan 2025
Nabin, J. A., et al.
(IEEE ICAECA 2025)

A Novel Temporal-Aware Adaptive Feature Selection Strategy for Network Intrusion Detection Systems Jan 2025
Nabin, J. A., et al.
(GINOTECH 2025)

Cross-Domain Adversarial Robustness and Explainability Stability in Phishing Detection Systems June 2026
Hassan, M. R., **Nabin, J. A.**, Toyee-E-Ferdoush
[10.1109/PECCII70991.2026.11661863](https://doi.org/10.1109/PECCII70991.2026.11661863) (PECCII 2026 (IEEE))

Unraveling the Factors of Procrastination: An Insightful Assessment of Procrastination Levels Through ML and XAI

Jan 2026

Sarker, M., Efat, A. H., **Nabin, J. A.**, Khan, M. A., Zibran, M. F.
(IEEE ICMI 2026)

Two-Stage Vision Transformer-Based Framework for Anomaly Detection and Classification in Surveillance Videos

Jan 2025

Nabin, J. A., et al.

(2025 International Conference on Electrical, Computer and Communication Engineering (ECCE))

Projects

FL-IDS Framework

2023 – Present

- CNN-LSTM federated learning architecture for privacy-preserving network intrusion detection
- Evaluated on NSL-KDD and CICIDS benchmark datasets across distributed clients

Adversarial Phishing Detection Study

2024 – 2026

- Measured SHAP explanation drift under PGD and C&W adversarial perturbations in phishing URL classifiers
- Quantified cross-domain robustness gaps; published at PECCII 2026 (IEEE)

Multimodal Meme Sentiment Analyzer

2024 – 2025

- Late-fusion architecture combining BERT text embeddings and ResNet image features
- SHAP applied to identify dominant modality per prediction

Skills

Programming Languages: Python, C, C++, SQL, JavaScript

ML / DL Frameworks: PyTorch, TensorFlow, Keras, Scikit-learn, Flower (Federated Learning)

Explainability & Security: SHAP, LIME, IBM Adversarial Robustness Toolbox (ART), Foolbox

Data & Visualization: Pandas, NumPy, Matplotlib, Seaborn, OpenCV

Environments & Tools: Git, GitHub, LaTeX, Jupyter Notebook, VS Code, Google Colab

Databases: MySQL, PostgreSQL, MongoDB

Awards

Research Publication Recognition: Miyan Publications Reward Ceremony 2026

Service

Conference Presenter: IEEE ICMI 2026, PECCII 2026, IEEE ICAECA 2025, GINOTECH 2025, ECCE 2025, ICCIT 2025, QPAIN 2025, Springer Conference 2024